

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors: Mid-Semester Progress Report

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Abstract—This report presents the mid-semester status of a classical (non-deep-learning) pipeline for texture-based fabric defect detection and localization on the AITEX Fabric Image Database (AFID). We have completed dataset indexing, mask matching, auditing, patch extraction, four texture descriptors (GLCM, LBP, Gabor, FFT), and a Mahalanobis-distance anomaly model with heatmap localization. A first worked example on three defective images gives a mean IoU of 0.083 (precision 0.094, recall 0.560). We treat this as a baseline and analyze its failure modes.

Index Terms—fabric defect detection, texture analysis, GLCM, local binary patterns, Gabor filters, AITEX AFID, classical computer vision

I. INTRODUCTION

Automated fabric inspection is a texture-analysis problem: a defect is a local deviation from a regular, periodic weave. This project models *normal* fabric texture and scores local patches by their statistical deviation from that model, using AFID [1], which provides grayscale fabric images with pixel-level ground-truth masks. Classical descriptors are used instead of learned features so the system stays interpretable and does not need large labeled sets, which suits AFID’s small defective-image count.

II. METHODOLOGY

The pipeline has five stages, run on Kaggle.

1) Dataset indexing. The dataset root is auto-discovered and files are bucketed into *normal*, *defect*, or *mask* by path keyword. Fabric ID and defect code are parsed from filenames; defective images are matched to masks by filename stem, with fallback suffix matching.

2) Preprocessing. CLAHE (clip limit 2.0, 8×8 tiles) followed by a mild Gaussian blur reduces illumination bias.

3) Patch extraction and labeling. 64×64 patches (stride 32 for the full audit, 64 for the worked example) are labeled defective if at least 10% of pixels overlap the ground-truth mask.

4) Texture descriptors. Each patch is described by GLCM statistics [2] (2 distances $\times$ 4 angles), uniform LBP [3] at two (P, R) settings, a Gabor bank [4] (2 frequencies $\times$ 4 orientations), and FFT radial-band energies (4 bands), giving a 68-dimensional vector.

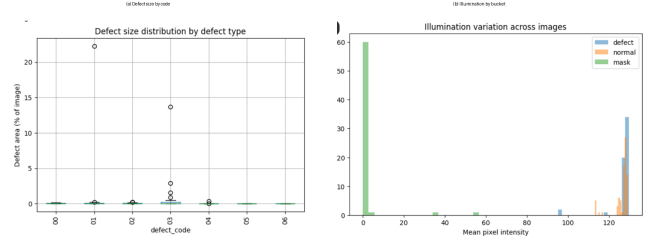


Fig. 1. (a) Defect area (% of image) by defect code. (b) Mean pixel intensity by file bucket; normal and defective images overlap near 120–130.

5) Adaptive anomaly model. Normal patches define a mean vector and regularized inverse covariance; each patch is scored by Mahalanobis distance. A per-fabric model is used when normal images exist for that fabric ID, otherwise a pooled model over all sampled normal textures is used. The threshold is the 95th percentile of pooled normal scores (9.547). Score maps are thresholded into a binary mask and evaluated by IoU, precision, and recall against ground truth.

III. RESULTS

Table I summarizes the indexed data and patch set. Figs. 1 and 2 show exploratory checks.

TABLE I
DATASET AND PATCH SUMMARY

Metric	Value
Image files	354 (141 normal / 106 defect / 107 mask)
Mask match rate	103/106 (97.2%)
Resolution	256×4096 , uint8
Total / normal / defective patches	71,120 / 70,645 / 475
Defective : normal ratio	1 : 149
GLCM / LBP / Gabor / FFT length	16 / 28 / 16 / 8 (total 68)

Worked example. The normal model was fit on 10 defect-free images (stride 64, 2560 patches). Three defective images were scored end to end. Table II and Fig. 3 give the results. All three used the pooled fallback model, because their fabric IDs have no defect-free counterpart.

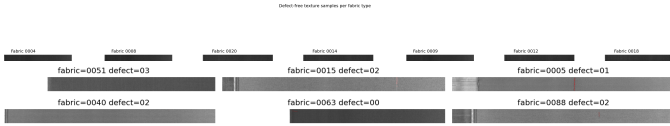


Fig. 2. Top: defect-free texture samples, one per fabric type. Bottom: ground-truth masks (red) overlaid on defective images. Defects are thin and small against a 256×4096 strip.

TABLE II
LOCALIZATION METRICS (POOLED MODEL)

Image	IoU	Precision	Recall
0099_019_02	0.000	0.000	0.000
0102_010_03	0.248	0.281	0.679
0087_030_02	0.000	0.000	1.000
Mean	0.083	0.094	0.560

IV. DISCUSSIONS

Mean recall (0.56) far exceeds mean precision (0.09). Image 0087_030_02 (recall 1.0, precision near zero) means almost the whole image was flagged, while 0099_019_02 is a missed defect. Image 0102_010_03 is the best case (IoU 0.248). A likely cause is that the pooled model mixes unrelated fabric textures, so one global threshold is too loose for some fabrics and too tight for others. With three images these numbers are a baseline, not a conclusion.

The data audit explains several design choices. Normal and defective intensities overlap almost completely (Fig. 1b), so intensity alone cannot separate them, and CLAHE mainly corrects image-to-image lighting. Most defects are small relative to a 64×64 patch (Figs. 1a and 2), which gives the 1:149 imbalance, so we evaluate with IoU/precision/recall rather than accuracy. Defect-free and defective images cover largely different fabric IDs (20 vs. 106; only 17 IDs have both), which limits per-fabric modeling and motivated the pooled fallback. Sampled images with up to 4 channels also mean the grayscale assumption needs an explicit check.

V. CONCLUSION

The project now has a working pipeline from indexing to heatmap-based localization and a first quantitative baseline (mean IoU 0.083). Planned work: (i) per-fabric thresholds and morphological post-processing to raise precision; (ii) evaluation on the full defective set rather than three images; (iii) descriptor ablation across GLCM, LBP, Gabor, and FFT; (iv) mapping numeric defect codes to AITEX’s named categories; and (v) handling the fabric-coverage and class-imbalance gaps.

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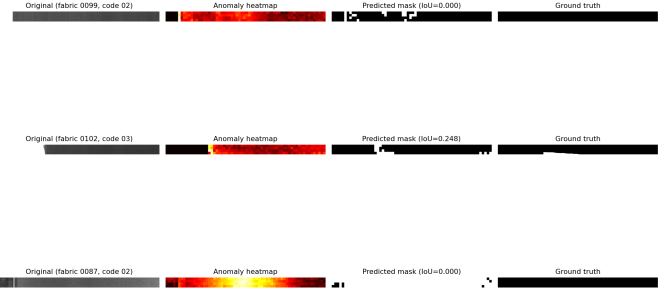


Fig. 3. Original, anomaly heatmap, predicted mask, and ground truth.

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